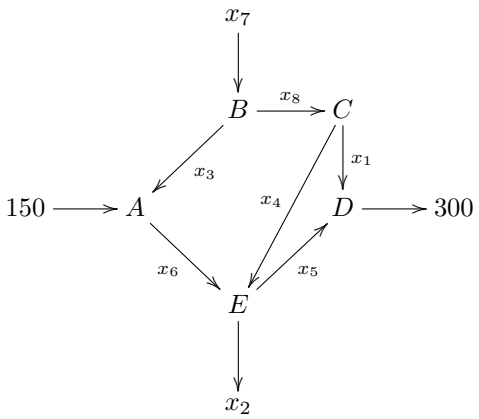
Examples of Network Flow Questions

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1. Consider the traffic flow described by the following diagram. The letters A through E label intersections. The arrows indicate the direction of flow (all roads are one-way) and their labels indicate flow in cars per minute.



(a) Write down a linear system describing the traffic flow, i.e., all constraints on the variables $x_i, i = 1, \ldots, 8$. (Do not solve the linear system at this stage.)

Solution.

A	$x_3 + 150$	$= x_6$
B	x_7	$= x_3 + x_8$
C	x_8	$= x_1 + x_4$
D	$x_1 + x_5$	$= 300$
E	$x_4 + x_6$	$= x_2 + x_5$
Total	$x_7 + 150$	$= x_2 + 300$

(b) The reduced row echelon form of the augmented matrix corresponding to the linear system in part (a) is:

$$\left[\begin{array}{cccccccc|c} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 300 \\ 0 & 1 & 0 & 0 & 0 & 0 & -1 & 0 & -150 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & -1 & -300 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & 150 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Find the general flow pattern (i.e., the general solution of the linear system).

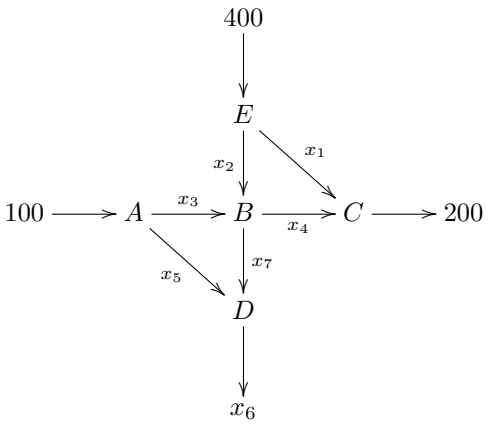
Solution. x_1, x_2, x_3, x_4, x_6: basic; x_5, x_7, x_8: free.

$$\begin{cases} x_1 = -x_5 + 300 \\ x_2 = x_7 - 150 \\ x_3 = x_7 - x_8 \\ x_4 = x_5 + x_8 - 300 \\ x_5 = \text{free} \\ x_6 = x_7 - x_8 + 150 \\ x_7 = \text{free} \\ x_8 = \text{free} \end{cases}$$

(c) Suppose we know that at least 170 cars per minute enter the intersection B from the north. Determine the minimum number of cars that will be leaving the intersection E towards south.

Solution. Since $x_2 = x_7 - 150$, from $x_7 \geq 170$ it follows that $x_2 \geq 170 - 150 = 20$.

2. Consider the traffic flow described by the following diagram. The letters A through E label intersections. The arrows indicate the direction of flow (all roads are one-way) and their labels indicate flow in cars per minute.



(a) Write down a linear system describing the traffic flow, i.e., all constraints on the variables $x_i, i = 1, \ldots, 7$. (Do not solve the linear system at this stage.)

Solution.

We can write the following six equations:

$100 = x_3 + x_5$

$x_2 + x_3 = x_4 + x_7$

$x_1 + x_4 = 200$

$x_5 + x_7 = x_6$

$400 = x_1 + x_2$

$100 + 400 = 200 + x_6$

(b) The reduced row echelon form of the augmented matrix corresponding to the linear system in part (a) is:

$$\left[\begin{array}{ccccccc|c} 1 & 0 & 0 & 1 & 0 & 0 & 0 & 200 \\ 0 & 1 & 0 & -1 & 0 & 0 & 0 & 200 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & -200 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 300 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 300 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Find the general flow pattern (i.e., the general solution of the linear system).

Solution. x_1, x_2, x_3, x_4, x_7 are basic and x_5, x_6 are free, and the general solution is:

$$\begin{cases} x_1 = 200 - x_4 \\ x_2 = 200 + x_4 \\ x_3 = x_7 - 200 \\ x_4 : \text{free} \\ x_5 = 300 - x_7 \\ x_6 = 300 \\ x_7 : \text{free} \end{cases}$$

(c) Suppose that due to road work, the flow along BC is limited to a maximum of 25 cars per minute. What is the maximum possible flow along EB?

Solution. Since $x_4 \leq 25$ and $x_2 = 200 + x_4$ we obtain $x_2 \leq 225$.